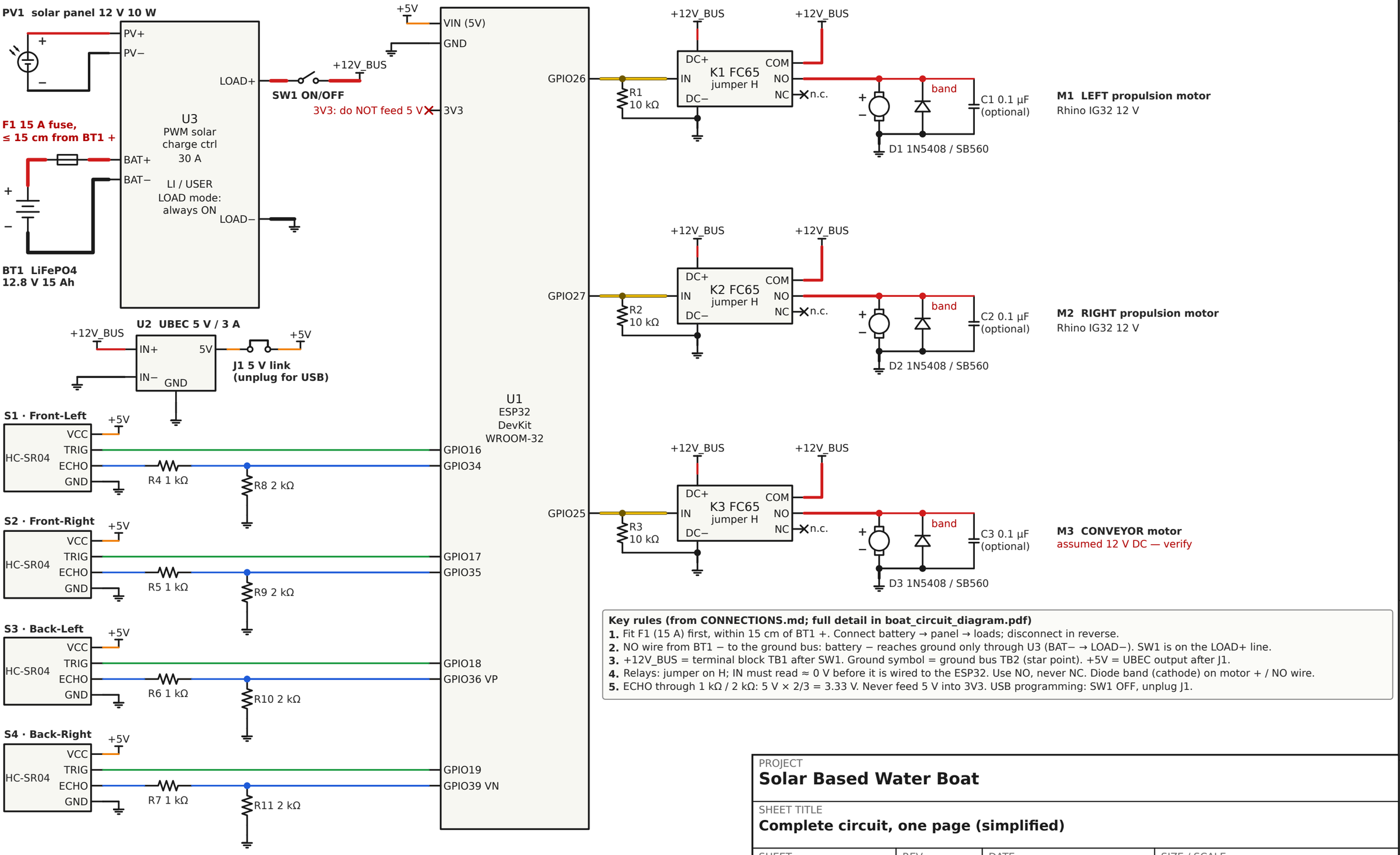
SOLAR BASED WATER BOAT — COMPLETE CIRCUIT (simplified, one page)



Key rules (from CONNECTIONS.md; full detail in boat_circuit_diagram.pdf)

1. Fit F1 (15 A) first, within 15 cm of BT1 +. Connect battery → panel → loads; disconnect in reverse.
2. NO wire from BT1 − to the ground bus: battery − reaches ground only through U3 (BAT− → LOAD−). SW1 is on the LOAD+ line.
3. +12V_BUS = terminal block TB1 after SW1. Ground symbol = ground bus TB2 (star point). +5V = UBEC output after J1.
4. Relays: jumper on H; IN must read ≈ 0 V before it is wired to the ESP32. Use NO, never NC. Diode band (cathode) on motor + / NO wire.
5. ECHO through 1 kΩ / 2 kΩ: 5 V × 2/3 = 3.33 V. Never feed 5 V into 3V3. USB programming: SW1 OFF, unplug J1.

Wires: +12 V · GND / − · +5 V · relay signal · TRIG · ECHO · ⏚ = ground bus TB2
Gauges: battery / main 12 V line 14 AWG · panel 18 AWG · COM, NO, motor 16–18 AWG · module DC± and UBEC input 20–22 AWG
5 V and logic ground 22 AWG · GPIO, TRIG, ECHO 22–24 AWG · ● = joined, no dot = not joined

PROJECT	**Solar Based Water Boat**
SHEET TITLE	**Complete circuit, one page (simplified)**
SHEET	**1 of 1**
REV	**A**
DATE	2026-10-24
SIZE / SCALE	A3 landscape · NTS

Source: CONNECTIONS.md
Generated by tools/make_schematic.py — edit the netlist there, then re-run